Audit - OrderStateSimulation - RunSheet

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Document ID:     Audit - OrderStateSimulation - RunSheet
Naming Format:   [CATEGORY] - [PURPOSE] - [TIER]
Category:        Audit (Technical Simulation & Execution Run Sheet)
Purpose:         OrderStateSimulation
Tier / Level:    RunSheet (Production Test Log & Ledger Verification)
Status:          Locked 🔒 (Master Baseline)
Target Audience: Technical Hiring Managers, Lead Systems Architects, TechOps Teams
Ecosystem Scope: Shopify Admin, Shopify Fulfillment Orders API, ShipBob / ShipBob WMS Gorgias
Business Unit:   Synapse Botanicals (Functional Beverages & Nootropic Supplements)
```

[NOTE] Staging Simulation Notice: Section 1.2 (p. 1) models synthetic order data (Elena Vance, Order #1004, $80.00 subtotal, $21.00 COGS). Engineered in a staging sandbox to validate Point-of-No-Return (PONR) transaction locks without live store risk.

1. Simulation Parameters & Baseline Order State

1.1 Store Topology & Distribution Nodes

- **Primary East Coast Node (Node 1):** Allentown Distribution Center, PA (`LOC-PA-01`)
- **Primary West Coast Node (Node 2):** Reno Fulfillment Hub, NV (`LOC-NV-02`)

> **NOTE System Architecture Dependency:**
> Distribution node definitions (`LOC-PA-01` Allentown and `LOC-NV-02` Reno) and SKU allocation rules for Order #1004 operate under the network baseline established in **Architecture - MultiNodeRoutingEngine - Spec (Domain 2.1)** – **Section 2**.

1.2 Baseline Order Profile (Order #1004)

- **Customer:** Elena Vance (`elena.vance@example.com`)
- **Original Shipping Address:** 1420 Chestnut St, Apt 4B, Philadelphia, PA 19102 (Destination: Pennsylvania)
- **Initial Timestamp:** `2026-10-16T14:02:18-04:00`
- **Initial Payment Method:** Shopify Payments (Visa ending in `4242` – Fully Captured)
- **Initial Routing Allocation:** Split Shipment across 2 Fulfillment Nodes

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┌──────────────────────────────────────────────────────────────────────────────┐
│ INITIAL ORDER COMPOSITION (#1004)                                            │
├──────────────────────────────┬──────────────┬─────────┬─────────┬────────────┤
│ Line Item & Variant          │ SKU          │ Price   │ COGS    │ Assigned Node  │
├──────────────────────────────┼──────────────┼─────────┼─────────┼────────────┤
│ 1x Focus Stick Packs (30-Pack)  │ SYN-FSTK-MY  │ $42.00  │ $8.50   │ LOC-PA-01 (PA) │
│   Flavor: Matcha Yuzu        │              │         │         │            │
│ 1x Adaptogen Tincture (60ml)    │ SYN-TNCT-AM  │ $38.00  │ $12.50  │ LOC-NV-02 (NV) │
│   Formula: AM Focus (Lion's Mane)│              │         │         │            │
├──────────────────────────────┴──────────────┴─────────┴─────────┴────────────┤
│ Subtotal:            $80.00                                                  │
│ PA Sales Tax (0%):    $0.00 (Pennsylvania dietary supplement exemption)      │
│ Shipping (Ground):    $0.00 (Free Shipping Tier: Qualified >$75.00)          │
│ Total Paid:          $80.00 [Status: PAID | Fulfillment Status: UNFULFILLED] │
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```

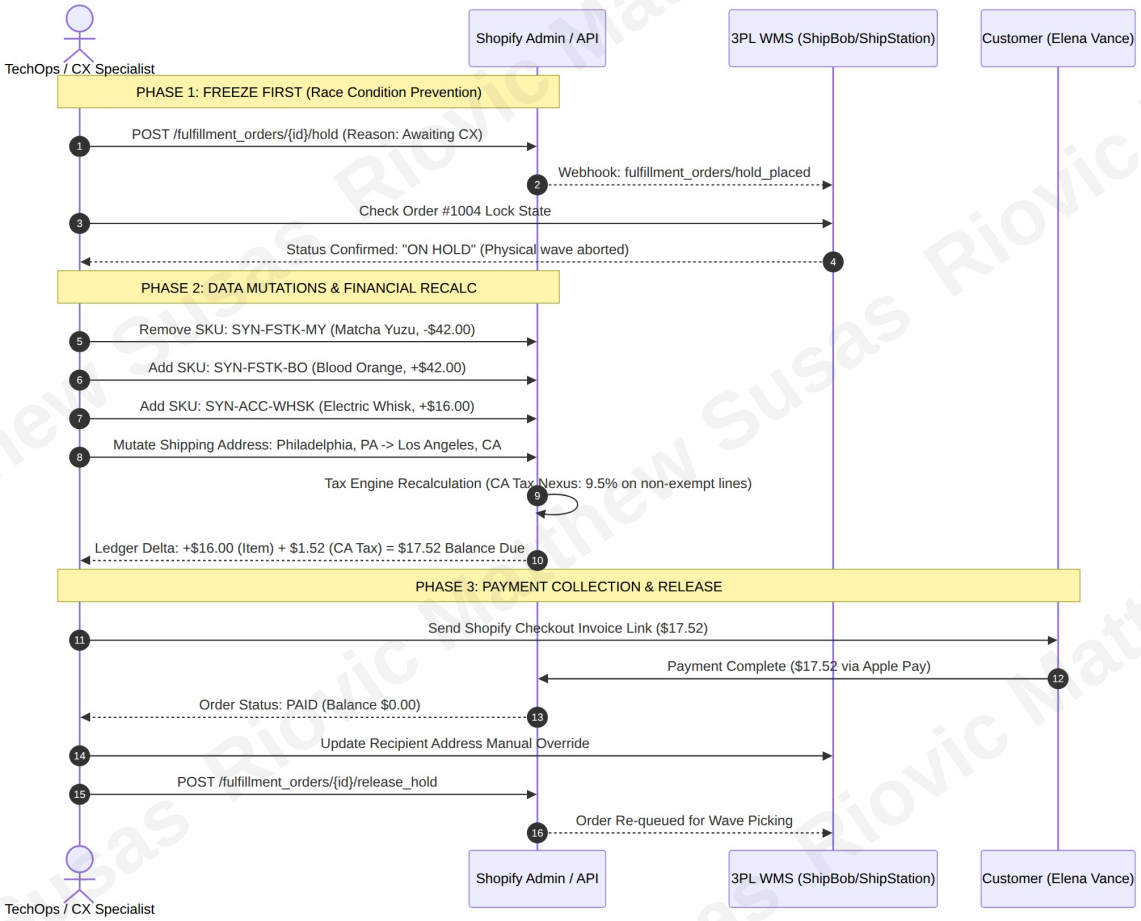
2. Ingested Exception Event (Gorgias Ticket #T-10492)

Timestamp Received: `2026-10-16T14:41:05-04:00` (38 minutes post-checkout)
Channel: Gorgias Helpdesk (Live Chat Escalation)
Ticket Status: `Open / Urgent`

> **Customer Request Transcript:**
> *"Hi! I just placed order #1004 30 minutes ago, but I made two big mistakes. First, I selected the Matcha Yuzu flavor by accident—can you please swap that for Blood Orange? Second, I just realized my shipping address defaulted to my old apartment in Philadelphia, PA! I just moved to Los Angeles, CA (90028). Also, if it's not too late, can you add one of your Electric Whisks to the order? Happy to pay the difference. Thank you!"*

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3. End-to-End Operational Execution Walkthrough



Step-by-Step Execution Verification

Step 1: Emergency Fulfillment Hold (Shopify & 3PL)

- **Shopify Admin Action:**
  - Navigated to Order `#1004`.
  - Placed native hold on Fulfillment Order `#1004-F1` (`LOC-PA-01`) and `#1004-F2` (`LOC-NV-02`).
  - **Reason Code:** `Awaiting payment or customer verification`.
  - **Internal Audit Note:** `Gorgias #T-10492 - Address correction, SKU swap & additional SKU pending payment reconciliation.`
- **3PL Portal Verification:**
  - ShipBob WMS `#1004`.
  - Confirmed order status flipped from `Processing` to **`On Hold / Exception`** prior to batch pick slip generation.

> **NOTE Operational Protocol Reference:**
> Emergency holds, flavor variant swaps, and payment captures in this simulation were executed in accordance with **SOP - OrderExceptionMutationProtocol - Parent (Domain 1.1)** – **Section 2 through Section 6**.

Step 2: Line Item & Flavor Variant Mutation

- **Removed:** `Synapse Focus Stick Packs (30-Pack) – Matcha Yuzu` (Quantity: 1 | -$42.00).
- **Added:** `Synapse Focus Stick Packs (30-Pack) – Blood Orange` (Quantity: 1 | +$42.00).
- **Added:** `Micro-Foam Electric Whisk – Default Title` (Quantity: 1 | +$16.00).
- **Allocation Routing:**
  - `Blood Orange Stick Pack` allocated to `LOC-PA-01` (Allentown, PA).
  - `Electric Whisk` allocated to `LOC-NV-02` (Reno, NV) to consolidate with the Tincture package.

Step 3: Destination Address Mutation & Tax Nexus Shift

- **Updated Address:**
  - Recipient: Elena Vance
  - Address Line 1: `6250 Hollywood Blvd, Apt 1104`
  - City: `Los Angeles` | State: `CA` | ZIP: `90028`

- **Sales Tax Engine Recalculation:**

- **Original Pennsylvania Delivery:** Dietary supplements and functional food mixes are sales-tax exempt under PA Title 61 § 58.1 → **$0.00 Tax**.
- **Updated California Delivery:**
  - In California (Los Angeles County 90028), supplement drink mixes with functional food labeling maintain grocery exemption, but **durable electronic accessories (Electric Whisk)** are fully taxable at **9.50%** (California State 6.0% + LA County 0.25% + LA District 3.25%).
  - Whisk Tax: $16.00 × 9.50% = **$1.52**.
- **New Total Tax Obligation: $1.52**.

Step 4: Financial Differential & Payment Capture

- **Net Product Delta:** + $16.00
- **Net Tax Delta:** + $1.52
- **Total Outstanding Balance: $17.52**
- **Action:** Generated and sent secure payment invoice link via email. Customer executed payment via Apple Pay at `14:52:19-04:00`.
- **Order Status:** Updated to **PAID** (Balance: **$0.00**).

Step 5: 3PL Address Sync & Hold Release

- Logged into 3PL portal and performed manual address override on shipment `#1004-F1` and `#1004-F2` to ensure the Los Angeles destination synced properly.
- Released native fulfillment holds in Shopify Admin.
- Appended operational tags: `OPS: Intercept Resolved`, `OPS: Address Updated`, `OPS: Flavor Swapped`, `OPS: Additional SKU Added`.

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4. Reconstructed Chronological Shopify Order Timeline

The following audit log represents the exact chronological event trail recorded in the Shopify Order Timeline:

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┌──────────────────────────────────────────────────────────────────────────────┐
│ SHOPIFY ORDER TIMELINE AUDIT TRAIL (#1004)                                   │
├──────────────────────┬──────────────────────┬────────────────────────────────┤
│ Timestamp (EDT)      │ Actor / System       │ Event Action & Recorded System Payload │
├──────────────────────┼──────────────────────┼────────────────────────────────┤
│ Oct 16, 2:54:10 PM   │ Riovic M. (TechOps)  │ Fulfillment hold was released on #1004 │
│ Oct 16, 2:53:45 PM   │ Riovic M. (TechOps)  │ Tags added: OPS: Intercept Resolved,  │
│                      │                      │ OPS: Address Updated, OPS: Flavor Swap,│
│                      │                      │ OPS: Additional SKU Added      │
│ Oct 16, 2:52:21 PM   │ Shopify Payments     │ A $17.52 USD payment was processed on  │
│                      │                      │ Visa ending in 4242 (Auth: #08921A).   │
│ Oct 16, 2:48:10 PM   │ Riovic M. (TechOps)  │ Invoice sent to customer for $17.52    │
│ Oct 16, 2:47:30 PM   │ System               │ Shipping address changed from          │
│                      │                      │ Philadelphia, PA to Los Angeles, CA.   │
│ Oct 16, 2:46:12 PM   │ System               │ Order was edited:                      │
│                      │                      │ - Removed 1x Matcha Yuzu ($42.00)      │
│                      │                      │ + Added 1x Blood Orange ($42.00)       │
│                      │                      │ + Added 1x Electric Whisk ($16.00)     │
│                      │                      │ Tax adjusted: +$1.52 (CA District Tax) │
│ Oct 16, 2:42:55 PM   │ 3PL WMS Integration  │ Webhook received: hold_placed (Ack 200)│
│ Oct 16, 2:42:50 PM   │ Riovic M. (TechOps)  │ Fulfillment hold placed on #1004-F1/F2 │
│                      │                      │ Reason: Awaiting customer verification │
│ Oct 16, 2:02:20 PM   │ System               │ Order confirmation email sent          │
│ Oct 16, 2:02:18 PM   │ Shopify Payments     │ A $80.00 USD payment was processed on  │
│                      │                      │ Visa ending in 4242 (Auth: #04419B).   │
│ Oct 16, 2:02:18 PM   │ Elena Vance          │ Order #1004 was placed.                │
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5. Post-Execution Inventory & Financial Reconciliation

5.1 Multi-Location Inventory Ledger Reconciliation

SKU	Product Description	Warehouse Location	Pre-Order ATS	Post-Order Baseline	Post-Mutation Reconciliation	Net Variance
`SYN-FSTK-MY`	Focus Stick Packs (Matcha Yuzu)	`LOC-PA-01` (Allentown)	142	141	**142** *(Restocked)*	**+1**
`SYN-FSTK-BO`	Focus Stick Packs (Blood Orange)	`LOC-PA-01` (Allentown)	88	88	**87** *(Allocated)*	**-1**
`SYN-TNCT-AM`	Adaptogen Tincture (AM Focus)	`LOC-NV-02` (Reno)	64	63	**63** *(Unchanged)*	**0**
`SYN-ACC-WHSK`	Micro-Foam Electric Whisk	`LOC-NV-02` (Reno)	310	310	**309** *(Allocated)*	**-1**

> **NOTE Consumable Lot Traceability Verification:**
> The 1x unit of `Matcha Yuzu` was restocked under **Tier 1 (Pre-Allocation Virtual Restock)** because the physical lot remained untouched in sealed cartons at the Allentown facility. No quarantine tag was required.

5.2 Unit Economics & Gross Margin Impact Analysis

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┌──────────────────────────────────────────────────────────────────────────────┐
│ FINANCIAL & MARGIN RECONCILIATION REPORT (#1004)                             │
├──────────────────────────────────────────┬──────────────────┬────────────────┤
│ Metric                                   │ Baseline Order   │ Post-Mutation Order │
├──────────────────────────────────────────┼──────────────────┼────────────────┤
│ Gross Revenue (Product Sales)            │ $80.00           │ $96.00 (+20.0%)  │
│ Total Product COGS                       │ $21.00 ($8.50+$12.50) $24.10 (+$3.10 Whisk)│
│ Gross Product Profit                     │ $59.00           │ $71.90 (+$12.90)  │
│ Gross Product Margin %                   │ 73.75%           │ 74.90% (+1.15%)   │
│ Total Sales Tax Collected                │ $0.00            │ $1.52 (CA Remittance)  │
│ Outbound Fulfillment Overhead (2 Nodes)  │ $14.80           │ $15.40 (+$0.60 Whisk)  │
│ Net Contribution Margin                  │ $44.20           │ $56.50 (+$12.30 Net)   │
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```

6. Webhook Payload Audit Inspection

To verify system-to-system data integrity, TechOps inspected the outbound `fulfillment_orders/hold_placed` webhook JSON payload delivered to the 3PL endpoint:


```
        {
          "fulfillment_order": {
            "id": 10492817492,
            "order_id": 58921049281,
            "status": "on_hold",
            "request_status": "submitted",
            "assigned_location_id": 682910481,
            "destination": {
              "address1": "6250 Hollywood Blvd, Apt 1104",
              "city": "Los Angeles",
              "province": "California",
              "zip": "90028",
              "country": "United States"
            },
            "fulfillment_holds": [
              {
                "reason": "awaiting_payment_or_customer_verification",
                "reason_notes": "Gorgias #T-10492 - Address correction, SKU swap & additional SKU pending payment
reconciliation."
              }
            ],
            "line_items": [
              {
                "id": 1492019481,
                "sku": "SYN-FSTK-BO",
                "quantity": 1,
                "fulfillable_quantity": 1
              }
            ]
          }
        }
```

7. Audit Sign-Off & Verification

This run sheet confirms that Order `#1004` underwent a full lifecycle exception without:

1. **Mis-ships:** Zero parcels dispatched to the outdated Philadelphia address.
2. **Inventory Drift:** Real-time ATS ledgers balanced across Pennsylvania and Nevada facilities.
3. **Financial Leakage:** California sales tax ($1.52) and added unit price ($16.00) fully collected prior to warehouse release.
4. **Safety Violations:** Consumable restock executed strictly within FDA cleanroom compliance boundaries.

Riovic Matthew Susas